

SAMARTH SAXENA

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SUMMARY

Electronics and Computer Engineering student at Thapar Institute with a strong foundation in C++, Python, Data Structures & Algorithms, and software development. Focused on building AI-powered applications using LangChain, Retrieval-Augmented Generation (RAG), multi-agent workflows, vector databases, and AWS deployment. Passionate about solving real-world engineering problems and developing scalable software systems through AI and modern software engineering practices.

EDUCATION

Thapar Institute of Engineering and Technology, Patiala

B.E. Electronics and Computer Engineering, CGPA: 8.52 Aug. 2023 – July 2027

Swaraj India Public School, Kanpur

Class XII (ISC Board) – 87% May 2023

PROJECTS

Multi Agent Research Assistant | Python, LangChain, Tavily API, BeautifulSoup, Streamlit, AWS | [View Project](#)

- Developed a Multi-Agent Research Assistant using LangChain, Mistral AI, Tavily Search, and BeautifulSoup to automate end-to-end research workflows.
- Designed specialized Search, Reader, Writer, and Critic agents to perform web retrieval, content extraction, report generation, and output evaluation.
- Integrated Tavily Search API with web scraping pipelines to collect and process real-time information from online sources.
- Leveraged LLMs and prompt engineering to generate structured research reports and provide AI-driven quality assessments.
- Deployed the application on AWS EC2 with a Streamlit frontend, enabling users to generate research reports through an interactive web interface.

AI Career Copilot | Python, LangChain, Mistral AI, FAISS, Streamlit, AWS EC2 | [View Project](#)

- Developed an AI-powered career assistant enabling resume analysis, job-description matching, interview preparation, and skill-gap identification.
- Built a Retrieval-Augmented Generation (RAG) pipeline using LangChain and FAISS to perform semantic search and contextual question answering over resumes and job descriptions.
- Integrated Mistral AI for personalized career recommendations, resume optimization suggestions, and interview guidance.
- Built and deployed an end-to-end AI workflow covering document ingestion, vector embedding generation, similarity search and AWS cloud deployment.

TECHNICAL SKILLS

Languages: C++, Python, SQL.

Core Concepts: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems.

AI & Data Science: Machine Learning, Generative AI, LangChain, RAG, Vector Databases, Mistral AI, OpenAI.

Tools & Cloud: AWS (EC2), Git, Streamlit, Jupyter Notebook, VS Code.

Soft Skills: Team Collaboration, Analytical Thinking, Problem Solving, Effective Communication.

CERTIFICATIONS & ACHIEVEMENTS

IBM Python for Data Science, AI & Development: Completed coursework covering Python for AI, data analysis, and machine learning applications.

Problem-Solving: Solved 200+ problems on platforms like LeetCode and GeeksforGeeks to strengthen algorithmic proficiency.

Extracurriculars: Contributed to technical workshops and participated in musical performances, requiring teamwork, coordination, and communication skills in high-pressure environments.